

IPL Match & Performance Analysis: A Data-Driven Study

Executive Summary

The Indian Premier League (IPL) is one of the most competitive and data-rich T20 cricket tournaments in the world. This project analyzes IPL match-level and ball-by-ball datasets to uncover patterns related to batting performance, bowling efficiency, toss decisions, venue influence, and match outcomes.

The datasets were cleaned, integrated, and transformed using Pandas and NumPy. Statistical analysis and visualizations were then performed using Matplotlib and Seaborn. The analysis revealed that batting performance has improved significantly over time, teams often gain an advantage while chasing, and death-over performance plays a crucial role in determining match outcomes.

The study demonstrates how data analytics can be used to identify trends, evaluate player performance, and support strategic decision-making in professional cricket.

Methodology

The project was completed in five major stages:

1. Data Collection and Understanding

Two datasets were used:

- **matches.csv** – Match-level information including season, venue, toss decisions, winners, and player-of-the-match awards.
- **deliveries.csv** – Ball-by-ball information including batter, bowler, runs scored, extras, wickets, and dismissal details.

2. Data Cleaning

The datasets were examined for missing values and inconsistencies.

The following preprocessing steps were performed:

- Missing city values were replaced with "Unknown".
- Missing player_of_match values were replaced with "No Award".
- Missing winner values were labeled as "No Result".

- Missing dismissal_kind values were replaced with "not out".
- Duplicate records were checked and removed if necessary.
- Date columns were converted into datetime format.

3. Data Integration

The match-level and delivery-level datasets were merged using the match identifier, creating a unified analytical dataset.

4. Feature Engineering

Several new variables were created:

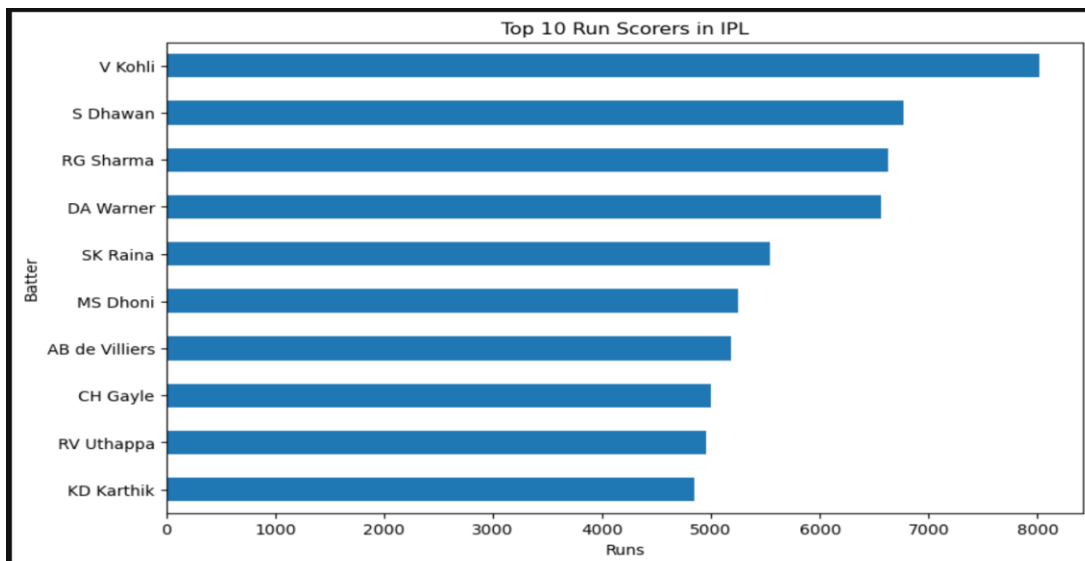
- Total Runs Scored
- Boundary Classification
- Over Phase Classification
- Year
- Month

5. Statistical Analysis and Visualization

Various aggregation techniques and visualizations were used to identify important IPL trends and performance patterns.

Key Findings

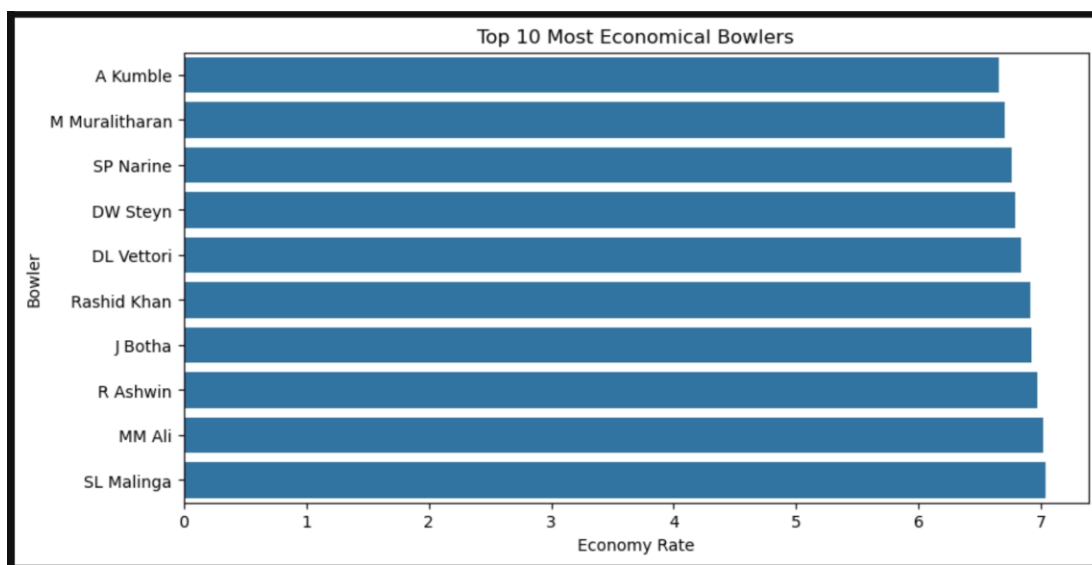
1. Virat Kohli is the Highest Run Scorer



- Virat Kohli emerged as the highest run scorer in IPL history, accumulating more than **8,000 runs** across multiple seasons.
- Other leading run scorers include **Shikhar Dhawan, Rohit Sharma, David Warner, and Suresh Raina**.
- These players have consistently performed across multiple IPL seasons and remain among the most successful batters in the tournament.
- The graph demonstrates that **long-term consistency and sustained performance** are the primary factors behind becoming one of the highest run scorers in IPL history.

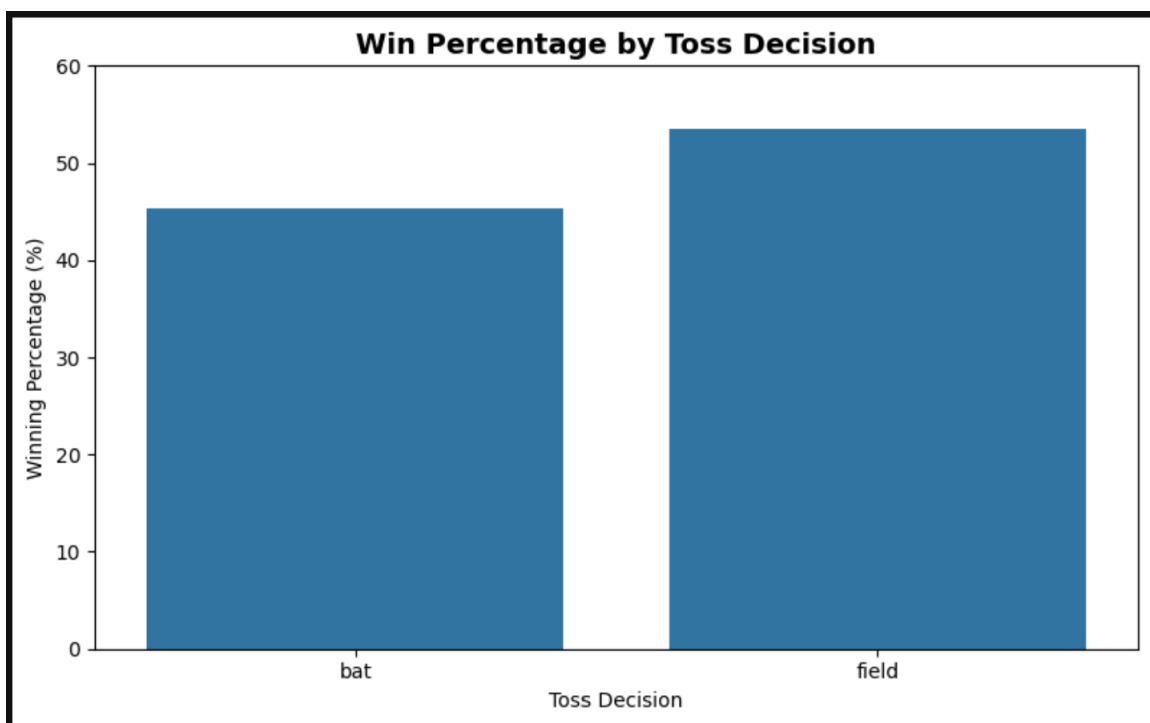
2. Economical Bowling Remains a Key Success Factor

	balls_bowled	runs_conceded	overs_bowled	economy_rate
bowler				
A Kumble	983	1089	163.833333	6.65
M Muralitharan	1581	1765	263.500000	6.70
SP Narine	4146	4672	691.000000	6.76
DW Steyn	2282	2583	380.333333	6.79
DL Vettori	785	894	130.833333	6.83
Rashid Khan	2901	3340	483.500000	6.91
J Botha	709	818	118.166667	6.92
R Ashwin	4679	5435	779.833333	6.97
MM Ali	770	900	128.333333	7.01
SL Malinga	2974	3486	495.666667	7.03



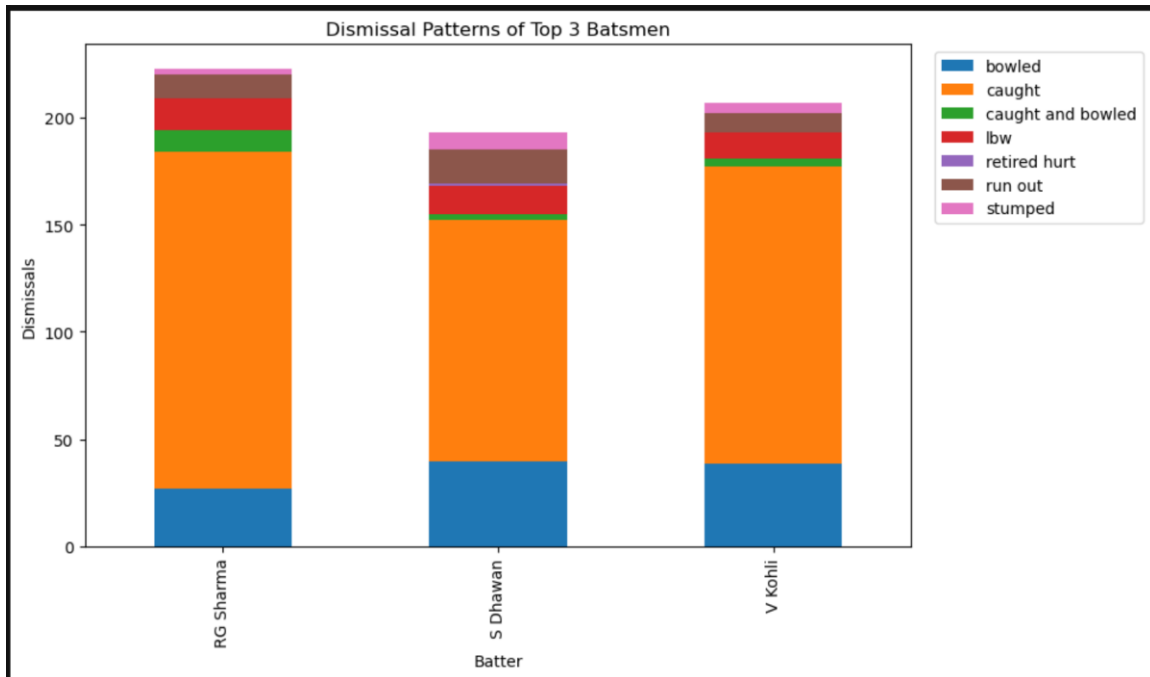
- Anil Kumble recorded the lowest economy rate among bowlers who delivered at least 500 balls in IPL history.
 - Other highly economical bowlers include M Muralitharan, Sunil Narine, Dale Steyn, and Daniel Vettori.
 - These bowlers consistently restricted opposition batters from scoring freely, helping their teams maintain pressure throughout the innings.
 - The analysis highlights the importance of economy rate as a key indicator of bowling effectiveness in T20 cricket.
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3. Toss Decisions Influence Match Outcomes



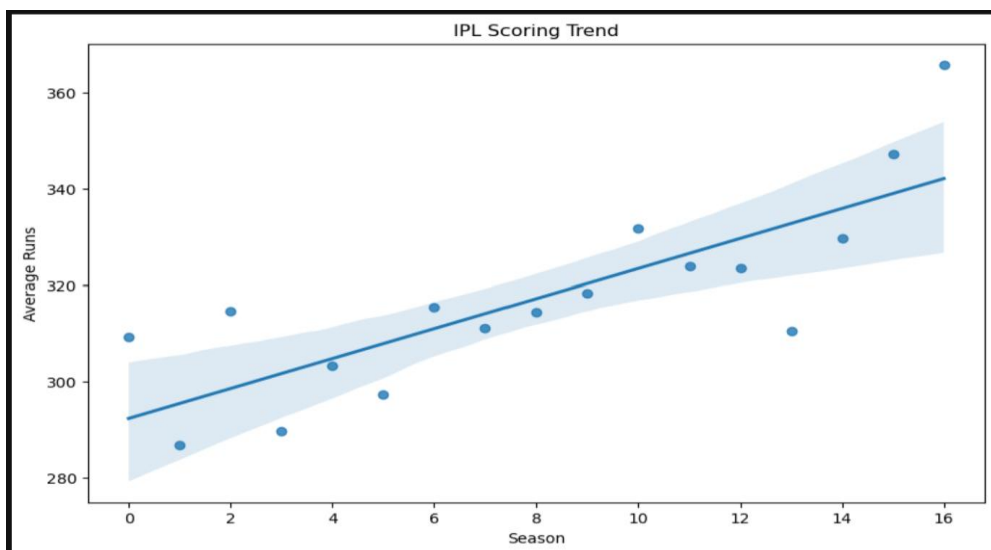
- Teams choosing to field first achieved a higher winning percentage than teams choosing to bat first.
 - Chasing teams benefit from knowing the target score before planning their innings.
 - The difference in win percentage suggests that toss decisions can influence match outcomes.
 - Fielding first appears to provide a slight strategic advantage in IPL matches.
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4. Dismissal Patterns of Top Batters



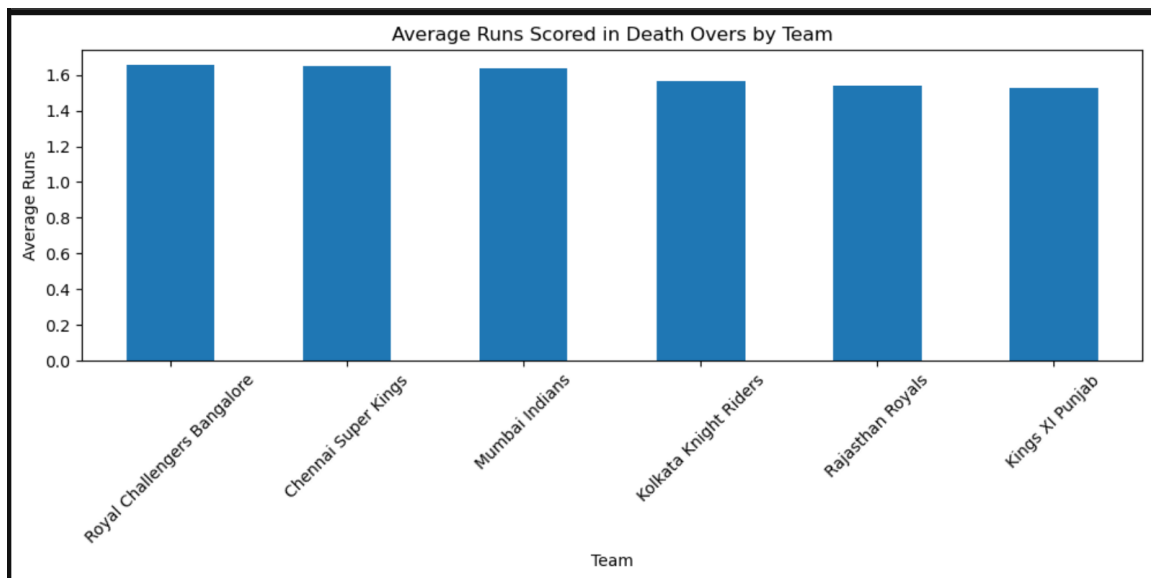
- The analysis focused on the dismissal patterns of Virat Kohli, Shikhar Dhawan, and Rohit Sharma, who are among the highest run scorers in IPL history.
- Caught dismissals emerged as the most common mode of dismissal for all three batters.
- Bowled, LBW, run out, and stumped dismissals occurred less frequently in comparison.
- The graph suggests that aggressive batting and attacking stroke play contribute significantly to wicket losses in T20 cricket

5. IPL Has Become More Batting Friendly



- The average runs scored per match have generally increased across IPL seasons.
- Early IPL seasons recorded average scores between **290 and 315 runs per match**, while recent seasons have exceeded **350 runs per match**.
- A noticeable rise in scoring can be observed after 2022, indicating a more aggressive and attacking batting approach.
- The upward regression trend line confirms a positive relationship between season progression and scoring rates.

6. Death Overs Significantly Impact Match Results



- Royal Challengers Bangalore, Chennai Super Kings, and Mumbai Indians recorded the highest average runs during death overs.
- The difference in average death-over scoring among teams is relatively small, indicating that most teams perform similarly during the final phase of the innings.
- Teams with stronger death-over scoring are better positioned to accelerate their run rate and maximize total scores.
- Effective finishing remains an important factor in both setting challenging targets and chasing difficult totals.

Strategic Recommendation

Based on the findings, IPL teams should prioritize two key strategies:

1. Prefer Chasing When Conditions Permit

The toss analysis indicates that teams often perform better while chasing. Captains should consider fielding first whenever pitch and weather conditions remain stable.

2. Invest in Death-Over Specialists

The final overs contribute significantly to overall match outcomes. Teams should focus on:

- Recruiting powerful finishers
- Developing specialist death-over bowlers
- Practicing high-pressure end-game scenarios

These improvements can directly influence winning percentages.

Limitations

Although the dataset provided valuable insights, several important factors were unavailable:

- Weather conditions
- Pitch reports
- Player fitness information
- Injury records
- Team combinations
- Advanced player tracking data

Including these variables could enable deeper analysis and more accurate predictive modeling.

Conclusion

This project successfully applied data science techniques to analyze IPL match and delivery data. Through data cleaning, feature engineering, aggregation, and visualization, several meaningful patterns were discovered.

The analysis demonstrated the dominance of elite batters, the value of economical bowling, the importance of toss decisions, and the growing significance of death-over performance. Furthermore, venue-based trends highlighted the potential impact of home advantage.

Overall, the project shows how data-driven approaches can provide actionable insights for teams, coaches, analysts, and cricket enthusiasts. The findings reinforce the growing role of analytics in modern cricket strategy and decision-making.